Department of
Computer Science and Engineering
Independent University, Bangladesh

Sec No.	CLOs	Marks Allocated	Marks Obtained	Course Code	CSE200
1	CLO1	40		**Course Title**	Computer Programming II
2				**Term**	Summer 2025
3				**Test**	Mid Term Examination
4				**Time**	90 Mins
5				**Date**	July 17, 2025
Total Marks		40		**Signature**	
Instructions		**1.** Answer ALL questions. **2.** Mobile phones or any electronic gadget is not allowed. **3.** Leaving room during the exam is strictly prohibited.			
Student Name				**Section**	
Student ID					

Section 1: Short Questions (4+2+4)

1. Define the difference between line **2 to 5** with line **6 to 7** by analyzing the code and associated output.

```
1 number = 7

2 if number % 2 == 0:
3   print(f"{number} is Even.")
4 else:
5   print(f"{number} is Odd.")

6 for i in range(1, 6):
7   print(i)
```

Write question 1 answer here:

2. Print the list in reverse order using slicing operator

```
gpa = [3, 3.5, 3.75, 4]]
```

Write question 2 answer here:

3. Draw a flow chart of the scenario mentioned**.**

A cinema charges different ticket prices based on the age of the customer and the day of the week**.** The pricing rules are as follows:

1. If the customer's age is less than 5, the ticket is free.
2. If the customer's age is between 5 and 18 (inclusive), the ticket costs $8.
3. If the customer's age is between 19 and 60 (inclusive):
    - If the day is Friday, the ticket costs $10 (discount day).
    - Otherwise, the ticket costs $12.
4. If the customer's age is above 60, the ticket costs $6 (senior citizen discount).

Write question 3 answer here:

Section 2: Output Tracing (4+4)

4.

```
def add_values(x, y):
    return x + y

def convert_and_add(s_number, base_value):
    num = int(s_number)
    return add_values(num, base_value)

def calculate_final_result(a, b):
    c = 23
    c += 30
    c = convert_and_add(b, a)
    final_result = add_values(c, a)
    return final_result

a = 87
b = '53'
result = calculate_final_result(a, b)
print(f"Value of c is: {result}")
```

Write question 4 output here:

5.

```
def display_list(lst, message):
    print(f"\n{message}: {lst}")

def main():

    items = ["apple", "banana", "cherry", "banana", "date"]
    display_list(items, "Original List")

    items.append("fig")
    items.insert(2, "grape")
    items.remove("banana")

    popped_item = items.pop()
    print(f”\nPopped item: {pooped item}”)

    idx = items.index("date")
    print(f"\nIndex of 'date' is: {idx}")

   items.sort()

   items.reverse()
   display_list(items, "After Reversing the List")

main()
```

Write question 5 output here:

Section 3: Error Tracing (6)

(**Underline the line has error and correct beside that**)

6.

```
1  def getItemQty():
2      qty = int('How Many item to purchase: '))
3      unitPrice = int('unit Price: ')
4      return [qty, unitPrice]
5  def calculatePrice (varArgs)
6      total = varArgs[0] * varArgs[1]
7      return
8   printAll(qty, uPrice, total):
9      print('QTY: {qty} \nUnit Price: {uPrice} \nPayable: {total}')
10 detail = getItemQty()
11 price = calculatePrice(detail)
12 printAll(detail[0],price)
```

Section 4: Code Refactoring (6)

7. Rewrite the program so that it can test at least 10 numbers and replace **for loop** using **while loop**.

```
print(“Prime number detection”)
def is_prime(n):
    if n < 2:
      return False
    for i in range(2, n//2 + 1):
      if n % i == 0:
        return False
    return True

prime_numbers = is_prime(87)
print("The result is:", prime_numbers)
```

Write question 7 answer here:

Section 5: Programming Exercise (4+6)

8. Suppose you have started working on an hourly basis as a SoD at IUB. You want to know how much BDT. you will earn a month given that you know how many hours you will work per month and how much you receive per hour. However, whatever amount you earn per month will have 25% of it deducted and that amount will be used to pay part of your tuition fees. Write a program that finds out the earnings before and after the deduction. Also find out the yearly earnings (without deduction) and how much of your tuition fees you are paying yourself annually. Assume that there are 22 days per month. Write a function named **calculateEarning()** that will calculate the monthly earning, earning after deduction, yearly earning and amount to be paid as tuition fees. **calculateEarning()** function should take daily working hours and earning per hour in BDT. as parameters.

Sample Input:
Enter the number of hours you will work per day:
Enter the amount you earn per hour:

Sample Output:
Monthly earning:
Monthly earning after deduction:
Yearly earning:
Amount paid as tuition fees per year:

9. Uber has recorded the fares of different rides throughout a day in a list. The list is populated with 5 items of float numbers standing for ride fares (in BDT**)**. As an Uber data analyst, you are asked to manage and analyze these fares using various list operations. Write a Python program that performs the following operations:

   (a) Add a new fare to the end of the list (e.g., for a newly completed ride).
   (b) Insert a promotional fare at the start of the list (e.g., a discounted short ride).
   (c) Remove a fare corresponding to a refunded ride.
   (d) Find and display the highest fare and the lowest fare from the list.
   (e) Calculate the total and average earning in the day.
   (f) Sort the fares in descending order to see the rides from highest to lowest fare.